Law of sines

1

a False

b False

c True

d True

2 No, because the law of sines requires one angle to find another angle.

3 Yes, because we can find the opposite angle to c using the angle sum of a triangle. Then we can find the other two sides using law of sines.

Unknown sides

4 $v = 10$

5

a $a = 10.5$

b $a = 6.9$

c $x = 13.1$

d $y = 4.4$

6

a $\angle A = 75.78^\circ$

b $b = 3.152 \text{ m}$

c $c = 5.723 \text{ m}$

7

a 29°

b $c = 37.11$

8

a

i $13 < a < 47$

ii $a = 17.97$

b

i $17 < a < 63.2$

ii $a = 27.64$

9

a 39°

b $k = 21.76$

Unknown angles

10

a $x = 30.4^\circ$

b $x = 32.7^\circ$

c $x = 26.0^\circ$

d $x = 43.2^\circ$

11

a $x = 133.13^\circ$

b $x = 149.12^\circ$

Ambiguous case

12

a 0

b 1

c 1

d 2

13

a No

b No

c Yes

d No

e Yes

f Yes

g No

h Yes

14

a No

b No

c Yes

d No

15

a $x = 41$

b 139°

Applications

16

a $a = 74^\circ$

b $d = 28.35 \text{ m}$

c $h = 16.66 \text{ m}$

17

a $\angle ABC = 21.89^\circ$

b $\angle BCA = 123.11^\circ$

c $AB = 29.2 \text{ km}$

18

a $\angle ADB = 23^\circ$

b $a = 35.65 \text{ yd}$

c $h = 20.4 \text{ yd}$

19

a $\angle ACB = 102^\circ$

b $b = 226.55 \text{ m}$

c $h = 117 \text{ m}$

20